

FRAME FOR PERFORMANCE

SSS 1-3 | Smart Irrigation Demonstrator

Frame and execute a research or engineering mission on feedback irrigation control. Produce quantitative evidence that can support a design or management decision.

MISSION KIT controller logs, sensors, calculator

EVIDENCE threshold, hysteresis, response or performance evidence

GM-471 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Frame with controller logs, sensors, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for threshold, hysteresis, response or performance evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Do not bypass safety controls. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Smart Irrigation Demonstrator, including a measurable acceptance criterion.

SMART RESEARCH MISSION

SSS 1-3 | Smart Irrigation Demonstrator

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MISSION KIT controller logs, sensors, calculator

EVIDENCE threshold, hysteresis, response or performance evidence

GM-472 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Design with controller logs, sensors, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for threshold, hysteresis, response or performance evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
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AGRISHINE TASK

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INSTRUMENT THE DESIGN

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MISSION KIT controller logs, sensors, calculator

EVIDENCE threshold, hysteresis, response or performance evidence

GM-473 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Instrument with controller logs, sensors, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for threshold, hysteresis, response or performance evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Do not bypass safety controls. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Smart Irrigation Demonstrator, including a measurable acceptance criterion.

CALIBRATE FOR PERFORMANCE

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MISSION KIT controller logs, sensors, calculator

EVIDENCE threshold, hysteresis, response or performance evidence

GM-474 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Calibrate with controller logs, sensors, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for threshold, hysteresis, response or performance evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Do not bypass safety controls. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Smart Irrigation Demonstrator, including a measurable acceptance criterion.

SMART RESEARCH MISSION

SSS 1-3 | Smart Irrigation Demonstrator

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MISSION KIT controller logs, sensors, calculator

EVIDENCE threshold, hysteresis, response or performance evidence

GM-475 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Model with controller logs, sensors, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for threshold, hysteresis, response or performance evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Do not bypass safety controls. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Smart Irrigation Demonstrator, including a measurable acceptance criterion.

VALIDATE THE DESIGN

SSS 1-3 | Smart Irrigation Demonstrator

Frame and execute a research or engineering mission on feedback irrigation control. Produce quantitative evidence that can support a design or management decision.

MISSION KIT controller logs, sensors, calculator

EVIDENCE threshold, hysteresis, response or performance evidence

GM-476 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Validate with controller logs, sensors, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for threshold, hysteresis, response or performance evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Do not bypass safety controls. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Smart Irrigation Demonstrator, including a measurable acceptance criterion.

EVALUATE FOR PERFORMANCE

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EVIDENCE threshold, hysteresis, response or performance evidence

GM-477 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Evaluate with controller logs, sensors, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for threshold, hysteresis, response or performance evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Do not bypass safety controls. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Smart Irrigation Demonstrator, including a measurable acceptance criterion.

SMART RESEARCH MISSION

SSS 1-3 | Smart Irrigation Demonstrator

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MISSION KIT controller logs, sensors, calculator

EVIDENCE threshold, hysteresis, response or performance evidence

GM-478 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Optimize with controller logs, sensors, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for threshold, hysteresis, response or performance evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
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AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Smart Irrigation Demonstrator, including a measurable acceptance criterion.

STRESS-TEST THE DESIGN

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MISSION KIT controller logs, sensors, calculator

EVIDENCE threshold, hysteresis, response or performance evidence

GM-479 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Stress-test with controller logs, sensors, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for threshold, hysteresis, response or performance evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Do not bypass safety controls. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Smart Irrigation Demonstrator, including a measurable acceptance criterion.

RECOMMEND FOR PERFORMANCE

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Frame and execute a research or engineering mission on feedback irrigation control. Produce quantitative evidence that can support a design or management decision.

MISSION KIT controller logs, sensors, calculator

EVIDENCE threshold, hysteresis, response or performance evidence

GM-480 - MISSION PATH

- DO** 1. Frame objective and performance metric. 2. Define system boundary, assumptions and constraints. 3. Recommend with controller logs, sensors, calculator. 4. Quantify uncertainty and quality-control checks. 5. Model/compare scenarios where appropriate. 6. Validate, evaluate and recommend.
- PROVE** Provide a traceable dataset, calculation/model/diagram, validation or QC note, and decision statement for threshold, hysteresis, response or performance evidence.
- SUCCESS** The method is reproducible, assumptions are explicit, uncertainty is addressed, and the recommendation is technically defensible.
- SAFETY** Do not bypass safety controls. Follow teacher instructions and school garden safety rules.

AGRISHINE TASK

Propose one optimization or next-stage prototype/test for the Smart Irrigation Demonstrator, including a measurable acceptance criterion.